

# Nicholas D. Gray, Ph.D.

## DATA SCIENTIST

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Data scientist with nearly four years of industry experience in insurance analytics, built on a decade of quantitative research training. Combines a Ph.D. in Cognitive Psychology with hands-on expertise in SQL, Python, and applied statistics to turn large, messy data sources into analyses, reports, and recommendations that stakeholders act on.

## TECHNICAL SKILLS

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|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages &amp; Tools</b>     | Python (advanced), SQL / Microsoft SQL Server, R, MATLAB, Microsoft Excel (advanced)                                                |
| <b>Modeling &amp; Statistics</b> | Regression (linear & logistic), XGBoost & gradient boosting (applied), Statistical inference, Experimental design, Model evaluation |
| <b>Data &amp; Analysis</b>       | ETL & data cleaning, Exploratory data analysis, Survey & qualitative analysis                                                       |
| <b>Communication</b>             | Stakeholder reporting & dashboards, Data visualization, Technical writing, Teaching & presentation                                  |

## PROFESSIONAL EXPERIENCE

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### Data Science Associate

Oct 2022 – Present

*Safety National Casualty Corporation · St. Louis, MO*

- Build analytics reports and dashboards used by internal teams and external stakeholders across the business.
- Extract and join data from large, complex sources in Microsoft SQL Server, then transform and clean it in Python for analysis.
- Run and adjust established XGBoost models, validating output and interpreting predictions for underwriting and claims questions.
- Design custom analyses out of exploratory investigation, tailoring method and reporting to the decision each stakeholder actually faces.

### Postdoctoral Researcher — Quantitative Analysis

Aug 2019 – Oct 2022

*Institute for Successful Longevity, Florida State University · Tallahassee, FL*

- Led statistical analysis for large-scale behavioral clinical trials on healthy cognitive aging, working across longitudinal datasets and multiple analysis platforms.
- Established statistical performance norms and built regression models identifying the variables that drive cognitive outcomes.
- Integrated qualitative streams — surveys and interviews — with quantitative measures to produce findings neither would support alone.
- Designed and programmed novel studies end to end and published the results in peer-reviewed journals.

### Graduate Researcher

Aug 2013 – May 2019

*Department of Psychology, Florida State University · Tallahassee, FL*

- Modeled experimental data with logistic regression and related statistical methods across several software platforms.
- Designed, programmed, and ran grant-funded experimental research; managed the lab, recruiting and supervising undergraduate researchers.
- Taught laboratory sections of Cognitive Psychology for three years and co-instructed an online Careers in Psychology course, translating research concepts and methods for non-expert audiences.

## SELECTED PROJECTS

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**adherence** – Python · [github.com/ndg13b/adherence](https://github.com/ndg13b/adherence)

Scores how regular a behavior is — not just how often it happens — from nothing but a list of timestamps, then forecasts when the next session will land. The same model applies to medication timing, exercise, physiotherapy, or a training program.

**CivicGateway** – JavaScript · [civicgateway.org](https://civicgateway.org)

A nonpartisan site showing upcoming elections and elected officials by city; static front end over a Supabase/PostgreSQL back end with a schema designed from scratch, row-level security, and an offline fallback.

**tuned-in** – JavaScript · [ndg13b.github.io/tuned-in](https://ndg13b.github.io/tuned-in)

A teaching collection pairing pop-culture clips with the cognitive psychology they demonstrate — and, just as often, get wrong. Each concept is labeled accurate, overstated, or myth.

## EDUCATION

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### Ph.D., Cognitive Psychology

2019

Florida State University

*Thesis: The causes of and influences on being reminded*

### M.S., Cognitive Psychology

2016

Florida State University

*Thesis: Avoiding interference: Differentiation and reminding*

### B.S., Psychology, cum laude

2012

Ohio State University

**Graduate methods training:** ANOVA, regression, meta-analysis, structural equation modeling

**Additional coursework:** Machine Learning with Python; Microsoft SQL Server (independently completed)

## SELECTED PUBLICATIONS

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Gray, N., & Charness, N. (2022). Technology obsolescence across the adult lifespan in a USA internet sample. *Frontiers in Public Health*, 10, 1005822.

Gray, N., Yoon, J.-S., Charness, N., Boot, W. R., et al. (2021). Relative effectiveness of general versus specific cognitive training for aging adults. *Psychology and Aging*, 37(2), 210–221.

Carr, D. C., Tian, S., He, Z., et al., incl. Gray, N. (2022). Motivation to engage in aging research: Are there typologies and predictors? *The Gerontologist*, gnac035.

Charness, N., Boot, W. R., & Gray, N. (2020). Mobile monitoring and intervention technology (MMI) for adaptive aging. In *Mobile Technology for Adaptive Aging: Proceedings of a Workshop* (pp. 21–40). National Academies Press.

*Conference presentations at the Psychonomic Society (2018) and the Cognitive Aging Conference (2022).*